Session 1: Fundamentals of Orbital Mechanics

Task 1: Derive the vis-viva equation from the energy integral

Follow the Newton law II, we have:

$$\boldsymbol{a} = \ddot{\boldsymbol{r}} = -\frac{\mu}{r^3}\boldsymbol{r}$$

$$\boldsymbol{v} \cdot \ddot{\boldsymbol{r}} = -\frac{\mu}{\boldsymbol{r^3}}\boldsymbol{r} \cdot \boldsymbol{v}$$

$$\boldsymbol{v} \cdot \ddot{\boldsymbol{r}} = -\frac{\mu}{\boldsymbol{r^3}}\boldsymbol{r} \cdot \dot{\boldsymbol{r}} \text{ (Because } \boldsymbol{v} = \dot{\boldsymbol{r}})$$

Know that:

- $\boldsymbol{v} \cdot \ddot{\boldsymbol{r}} = \frac{d}{dt}\left(\frac{1}{2}\boldsymbol{v} \cdot \boldsymbol{v}\right) = \frac{d}{dt}\left(\frac{1}{2}v^2\right)$
- $\boldsymbol{r} \cdot \dot{\boldsymbol{r}} = r\dot{r}$**,** so that: $-\frac{\mu}{r^3}(\boldsymbol{r} \cdot \dot{\boldsymbol{r}}) = -\frac{\mu\dot{r}}{r^2} = \frac{d}{dt}\left(\frac{\mu}{r}\right)$

Substitute into the equation:

$$\frac{d}{dt}\left(\frac{1}{2}v^2\right) = \frac{d}{dt}\left(\frac{\mu}{r}\right) => \frac{d}{dt}\left(\frac{1}{2}v^2 - \frac{\mu}{r}\right) = 0$$

Intergrating both sides with respect to time, we obtain the specific mechanical energy ϵ as a conserved constant:

$$\epsilon = \frac{1}{2}v^2 - \frac{\mu}{r} \qquad (1)$$

At the periapsis $r = r_p$ and apoapsis $r = r_a$, the velocity vector is perpendicular to the radius vector $\boldsymbol{v} \perp \boldsymbol{r}$. Therefore, the specific angilar momentum $h = \|\boldsymbol{r} \times \boldsymbol{v}\|$ as:

$$h = r_p v_p = r_a v_a => v_a = \frac{v_p r_p}{r_a} \qquad (2)$$

Apply equation (1) at periapsis and apoapsis:

$$\epsilon = \frac{1}{2}v_p^2 - \frac{\mu}{r_p} = \frac{1}{2}v_a^2 - \frac{\mu}{r_a}$$

Substitute v_a from (2) into the energy equation above:

$$\frac{1}{2}v_p^2 - \frac{\mu}{r_p} = \frac{1}{2}\left(v_p\frac{r_p}{r_a}\right)^2 - \frac{\mu}{r_a}$$

$$\Rightarrow \frac{1}{2}v_p^2\left(\frac{r_a^2 - r_p^2}{r_a^2}\right) = \mu\left(\frac{r_a - r_p}{r_p r_a}\right)$$

Divide both sides by $r_a - r_p$ and simplify:

$$\Rightarrow v_p^2 = \frac{2\mu r_a}{r_p(r_a + r_p)}$$

Substitute v_p^2 back into the energy equation (1) at the periapsis:

$$\epsilon = -\frac{\mu}{r_a + r_p} = -\frac{\mu}{2a}$$

Task 2: Summary of the six elements

- Semi-major axis: Defines the ssize of the orbit and the specific mechanical energy of the satellite.
- Eccentricity: Defines the geometric shape of the orbit ($e = 0$: circular, $0 < e < 1$: eleptical, $e = 1$: parabolic, $e > 1$: hyperbolic)
- Inclination: The tilt angle between the orbital plane and Earth's equatiorial plane $0 \leq i \leq 180$
- Right Ascension of the Ascending Node: The orientation angle of the orbital plane in space, measured from the Vernal Equinox to the ascending node.
- Argument of Perigee: The angle within the orbital plane measured counterclockwise from the ascending node to the perigee.
- True Anomaly: The angle specifying the actual position of the satellite along its orbit at time t, measured from the perigee.

Task 3: For an orbit with $r_p = 500\ km$ altitude and $r_a = 35786\ km$ altitude: find a,e, the period, and the speeds at periapsis and apoapsis.

- Constant & Reference Parameters:
  - Mean equatorial radius: $R_E = 6378\ km$
  - Standard gravitational parameter: $\mu = 398600\ km^3/s^2$
- Given Input Parameters:

- Periapsis altitude: $h_p = 500\ km \Rightarrow r_p = 6378 + 500 = 6878\ km$
- Apoapsis altitude: $h_a = 35786\ km \Rightarrow r_a = 6378 + 35786 = 42164\ km$

Step 1: Find semi-major axis:

$$a = \frac{r_p + r_a}{2} = \frac{6878 + 42164}{2} = 24521\ km$$

Step 2: Find eccentricity:

$$e = \frac{r_a - r_p}{r_a + r_p} = \frac{42164 - 6878}{42164 + 6878} \approx 0.7195$$

Step 3: Find orbital period:

$$T = 2\pi\sqrt{\frac{a^3}{\mu}} = 2\pi\sqrt{\frac{24521^3}{398600}} \approx 38213.647\ s$$

Step 4: Find speed at periapsis:

$$v_p = \sqrt{\mu\left(\frac{2}{r_p} - \frac{1}{a}\right)} = \sqrt{398600\left(\frac{2}{6848} - \frac{1}{24521}\right)} \approx 10.008\ km/s$$

Step 5: Find speed at apoapsis:

$$v_a = \frac{v_p r_p}{r_a} \approx \frac{10.008 \cdot 6878}{42164} \approx 1.631\ kms$$

Task 4: Python code for Example 1, 2, 3

```
import numpy as np
from astropy import units as u
from astropy.constants import R_earth
from poliastro.bodies import Earth
from poliastro.twobody import Orbit

# Overriding Earth.k with course reference constants for exact matching
# R_E = 6,378 km, mu = 398,600 km^3/s^2
MU_EARTH = 398600.0 * (u.km**3 / u.s**2)
R_E = 6378.0 * u.km

# ======================================================================
# PART 1: REPRODUCE ALL THREE WORKED EXAMPLES
# ======================================================================

print("--- WORKED EXAMPLE 1: Period of LEO (ISS Altitude, 420 km) ---")
h1 = 420.0 * u.km
a1 = R_E + h1  # 6,798 km
```

```
# Orbit creation using poliastro
orb1 = Orbit.from_classical(
    Earth,
    a=a1,
    ecc=0.0 * u.one,
    inc=0.0 * u.deg,
    raan=0.0 * u.deg,
    argp=0.0 * u.deg,
    nu=0.0 * u.deg,
)

# Period using Kepler's Third Law: T = 2*pi*sqrt(a^3 / mu)
T1 = 2 * np.pi * np.sqrt(a1**3 / MU_EARTH)
v1 = np.sqrt(MU_EARTH / a1)

print(f"Semi-major axis (a): {a1:.0f}")
print(f"Orbital Period (T): {T1.to(u.s):.1f} ({T1.to(u.min):.2f})")
print(f"Orbital Speed (v):  {v1.to(u.km/u.s):.2f}\n")


print("--- WORKED EXAMPLE 2: Circular Speed at 400 km ---")
h2 = 400.0 * u.km
r2 = R_E + h2  # 6,778 km

# Circular speed formula: v = sqrt(mu / r)
v2 = np.sqrt(MU_EARTH / r2)

print(f"Orbital Radius (r): {r2:.0f}")
print(f"Circular Speed (v): {v2.to(u.km/u.s):.2f}\n")


print("--- WORKED EXAMPLE 3: Radius of Geostationary Orbit ---")
T3 = 86164.0 * u.s  # One sidereal day

# Invert Kepler's 3rd law: a = (mu * T^2 / (4 * pi^2))^(1/3)
a3 = ((MU_EARTH * T3**2) / (4 * np.pi**2)) ** (1 / 3)
h3 = a3 - R_E
v3 = np.sqrt(MU_EARTH / a3)

print(f"Semi-major axis (a): {a3.to(u.km):.0f}")
print(f"GEO Altitude (h):   {h3.to(u.km):.0f}")
print(f"Orbital Speed (v):  {v3.to(u.km/u.s):.2f}\n")


# ======================================================================
# PART 2: KEPLER SOLVER (NEWTON-RAPHSON)
# ======================================================================


def solve_kepler(M_rad: float, e: float, tol: float = 1e-12) -> float:
    """Solves Kepler's Equation M = E - e*sin(E) using Newton-Raphson method

    and computes the True Anomaly (nu) in degrees.
    """
    # 1. Newton-Raphson iteration for Eccentric Anomaly (E)
    E = M_rad if e < 0.8 else np.pi

    while True:
        f = E - e * np.sin(E) - M_rad
        f_prime = 1.0 - e * np.cos(E)
        delta = f / f_prime
        E -= delta
        if abs(delta) < tol:
            break

    # 2. Convert Eccentric Anomaly (E) to True Anomaly (nu)
    tan_nu_2 = np.sqrt((1 + e) / (1 - e)) * np.tan(E / 2.0)
    nu_rad = 2.0 * np.arctan(tan_nu_2)

    # Convert to degrees and map to [0, 360)
    nu_deg = np.degrees(nu_rad) % 360.0
```

```
    return nu_deg


print("--- KEPLER SOLVER VERIFICATION ---")
M_test = 1.0  # radians
e_test = 0.35

nu_calc = solve_kepler(M_test, e_test)
print(f"Inputs:   M = {M_test} rad, e = {e_test}")
print(f"Output:   nu = {nu_calc:.4f} degrees")
print(f"Expected: nu ~ 97.6 degrees")
```

Task 5:

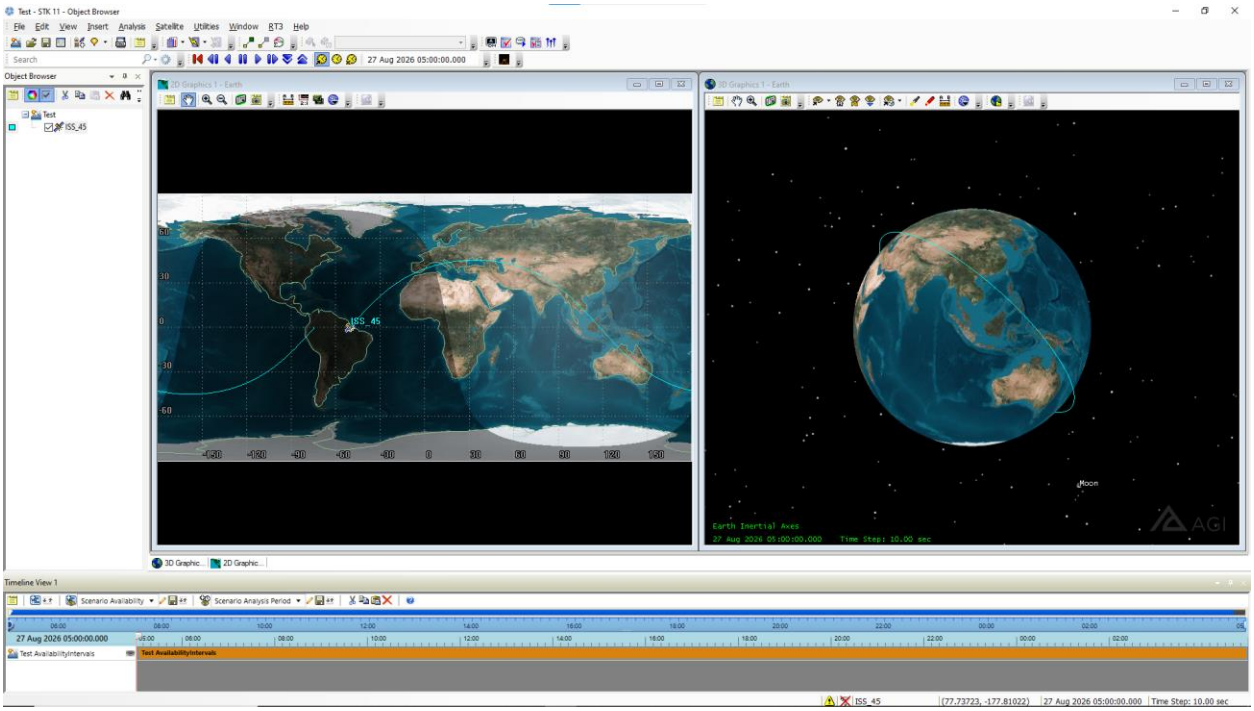


Figure 1: Inclination = 45

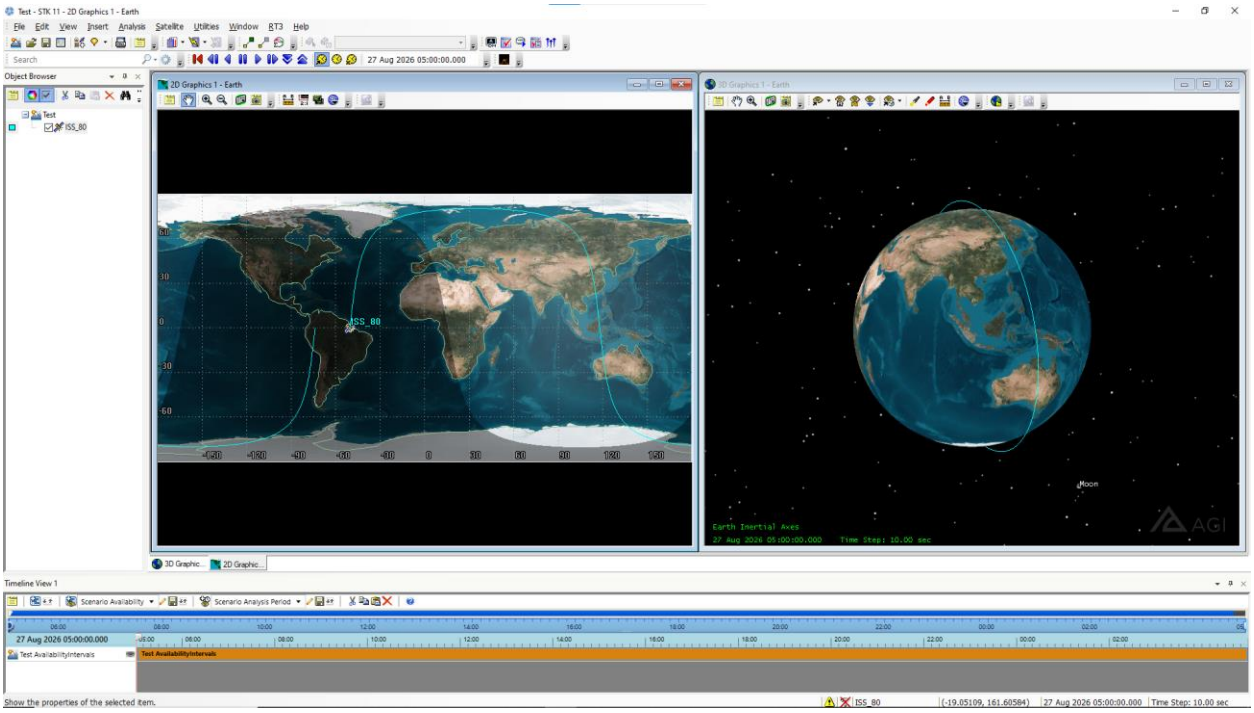


Figure 2: Inclination = 80